

# ATIQUE AHMAD

Machine Learning & Full-Stack Software Engineer

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## Summary

Software engineer with 3+ years of experience across machine learning, LLMs and full-stack development. I train and deploy production ML models, build RAG and agentic AI systems, and ship end-to-end products with Python, Go, React and cloud infrastructure on AWS, GCP and Azure, backed by an MS in Data Science.

## Experience

**Software Engineer, ML/AI** | *Paradigm Networks, Inc.*

**Jan 2023 – Present**

- **Factuality model:** Owned the full pipeline for a RAG factuality detector: curated 10K in-house samples, fine-tuned IBM Granite Guardian 3.1 (2B) with rsLoRA ( $r=256$ ), a custom prompt-loss-weighted SFT trainer and class-balanced sampling on RunPod A40 GPUs, then merged the adapter and published weights to S3. Delivered **91.7% accuracy and 86.7% F1**, beating GPT-5 (86.3% accuracy) and Claude 3 Sonnet (74.9%).
- **Hallucination model:** Co-developed a two-phase fine-tuned Granite detector (64K open-source + 10K in-house samples) that reached **98.1% F1 and 97.5% accuracy**, ahead of GPT-5 (88.2% accuracy) and Gemini 2.0 Flash (85.0%); owned its production serving and upgrades.
- **Model serving & deployment:** Deployed both detectors as GPU microservices: vLLM's OpenAI-compatible server behind async FastAPI (logprob-based confidence, warm-up, connection pooling), packaged as multi-stage distroless Docker images with weights from S3, and shipped via CodeBuild/ECR and GitHub Actions to AWS EKS.
- **Agentic evaluation platform:** Architected and built the service (FastAPI, MongoDB, DeepEval G-Eval) that scores every AI-agent run from its trace across 9 metrics (task success, plan quality, workflow completion, hallucination, toxicity); added local judges matching GPT-5 on 89% of plan-quality verdicts, integrated across the Go control server, LangGraph platform and React web app.
- **RAG verification APIs:** Built chunk-level RAG response verification, answer relevancy (DeepEval **0.90 F1** vs. RAGAS 0.85) and DeepEval faithfulness checks, with multi-cloud LLM support across AWS Bedrock (Claude, Llama 2), GCP Vertex AI (Gemini), Azure OpenAI (managed identity) and OpenAI with provider fallback.
- **PR review agent:** Built a LangGraph code-review agent that analyses pull-request diffs and publishes findings, traced with LangSmith and benchmarked against CodeRabbit.
- **MLOps:** Built CI test pipelines (pytest, mocked MongoDB) and hardened ML containers against vulnerabilities.

**Teaching Assistant / Lab Engineer** | *ITU Lahore — mentored 100+ students*

**Jan 2024 – Jul 2024**

**Software Engineer Intern** | *Clicky — full-stack web development*

**Jul 2022 – Oct 2022**

## Research & Projects

**Retrieval-Augmented Reasoning for Small LLMs** | *vLLM, FAISS, nomic/jina embeddings, Qwen3*

MSDS research

- Built a retrieval pipeline that turns Qwen3-8B teacher solutions (served with vLLM) into an embedding memory bank and a 600K-edge reasoning graph, then injects retrieved reasoning into a Qwen3-1.7B student.
- Gated retrieval raised held-out accuracy from **75.0% to 85.4% (+10.4 pp)**; step-level retrieval lifted MATH-500 from 70% to 76%. Reproduced Retrieval-of-Thought (arXiv:2509.21743) at +6.2 pp and proved through ablations that the injection site, not the templates, drives its failure mode.

## Education

**M.S. Data Science** | *Information Technology University, Lahore*

**2025 – Present**

**B.S. Computer Engineering** | *Information Technology University, Lahore* / CGPA 3.23

**2019 – 2023**

## Skills

|                             |                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------|
| <b>Machine Learning:</b>    | Deep learning, LLM fine-tuning (LoRA/PEFT), NLP, computer vision, self-supervised learning    |
| <b>GenAI &amp; Agents:</b>  | RAG, AI agents, prompt engineering, LLM evaluation, LangChain, LangGraph, LangSmith, DeepEval |
| <b>ML &amp; Data Tools:</b> | PyTorch, Hugging Face, scikit-learn, XGBoost, FAISS, Pandas, NumPy, Tableau, Power BI         |
| <b>Backend:</b>             | Python, Go, FastAPI, Django REST Framework, REST APIs, MongoDB, PostgreSQL, SQL               |
| <b>Frontend:</b>            | React, TypeScript, JavaScript, Next.js, Tailwind CSS, HTML/CSS                                |
| <b>MLOps &amp; Cloud:</b>   | vLLM, Docker, Kubernetes (EKS), Helm, CI/CD (GitHub Actions), AWS, GCP, Azure, RunPod         |